

Technical Report: Mobile-Money Credit Risk Modeling

Prepared for: Head of Credit Risk

Executive Summary

This report evaluates the quality, reliability and reproducibility of a machine learning solution for mobile-money credit risk. The analysis focuses on four areas: data quality, feature engineering, prevention of data leakage and reproducible model development. The data quality audit identified ten columns requiring attention based on missingness, validity checks, duplication and inconsistent formats. Remediation was selected according to the evidence observed during analysis. Feature engineering created a feature set which follows the five domain families: *RFM*, *ratios*, *velocity/trend*, *stability* and *behavioral features*. All historical aggregates follow the as-of rule, whereby information used at a scoring time t must be available strictly before t . Two intentionally planted leakage sources were also evaluated. Their impact was quantified by comparing cross-validated ROC-AUC before and after removal. The final solution used a single Scikit-learn pipeline with preprocessing, model fitting and evaluation performed without training information crossing validation folds. The final model which was leakage free achieved a mean group aware cross-validated ROC-AUC of **0.6250** and PR-AUC as **0.1349**. The resulting pipeline was serialized with `joblib` and saved for reload.

1 Data Quality Audit

The assessment combined descriptive statistics and validation checks. Duplicate records and invalid representations are resolved before feature construction, while treatment of missing values is performed as part of the modeling pipeline. Numerical missingness in GPS is retained through a missingness indicator. Raw identifiers that do not represent generalizable behavior are excluded from the feature matrix. The variables identified as becoming available after the intended scoring decision, including `manual_review_score` and `settlement_status`, are excluded from the model. These controls ensure that the final feature set is based on information that is valid, reproducible and available at the time of credit-risk decision making.

Table 1 presents the ten variables with data quality or model reliability concerns. The remediation strategies were selected to preserve usable observations while preventing invalid values and future information from entering the modeling pipeline.

Table 1: Data quality audit and selected remediation strategies

Column	Issue	Evidence	Missingness	Remediation
<code>txn_id</code>	Duplicate transaction records	Approximately 3,600 exact duplicate records were identified.	–	Remove exact duplicate records before feature construction. Retain the transaction identifier for traceability and lineage.
<code>amount</code>	Inconsistent numeric representation	Transaction amounts contain representations such as <code>'71/-Dr'</code> , <code>'4,742/- Dr'</code> , <code>'1,701/-'</code> and the column is of type <code>str</code> .	–	Convert to a consistent numeric representation and validate the resulting values before feature construction.
<code>txn_time</code>	Inconsistent timestamp formats	Multiple timestamp representations were encountered, including <code>YYYY-MM-DD HH:MM:SS</code> and <code>DD/MM/YYYY HH:MM</code> .	–	Parse supported formats explicitly and convert valid timestamps to a common datetime representation.
<code>gps_lat</code> and <code>gps_lon</code>	High missingness	Approximately 36.21% of coordinates are missing. When compared with region, missingness is at approximately 35.4–36.7%, indicating that <code>region</code> does not explain the missingness. Domain knowledge indicates that GPS/network coverage may depend on underlying location.	MNAR	The raw coordinates are excluded from the final feature set. Instead, the derived <code>gps_missing</code> indicator is retained to capture information about GPS availability.

Column	Issue	Evidence	Missingness	Remediation
<code>agent_id</code>	Structured missingness	Approximately 13.69% are missing. Missingness is 45.2% in Arusha, 45.1% in Jinja and 44.8% in Mwanza, compared with 0% in all other regions. The pattern is associated with the observed <code>region</code> field.	MAR	Exclude from the predictive feature set because it is an identifier.
<code>device_id</code>	Missing identifier values	Approximately 5.91% are missing. Regional missingness is approximately uniform, ranging from 5.6–6.1%. Fraud rates are essentially unchanged when the identifier is missing (8.38%) versus present (8.40%).	MCAR	Exclude from the predictive feature set because it is not part of the approved behavioral feature set. No modeling imputation is required.
<code>reg_id</code>	Identifier / high-cardinality risk	The field functions as an identifier rather than a stable behavioral measure.	–	Exclude from the predictive feature set. Retain outside the feature matrix for record tracking and data lineage.
<code>manual_review_score</code>	Potential target leakage	The variable is associated with the manual-review process and may contain information generated after a transaction enters the review workflow.	–	Exclude from the predictive feature set and enforce an explicit as-of/feature availability rule during feature development.
<code>settlement_status</code>	Post-outcome information	Settlement status becomes known during subsequent transaction processing. Using it at the original scoring timestamp would expose the model to future information.	–	Exclude from the predictive feature set and enforce temporal availability checks so that only information available before scoring is used.

2 Feature Dictionary

The engineered features were designed according to the five domain feature families. Historical aggregates used windows ending strictly before the scoring timestamp t .

Table 2: Engineered feature dictionary

Feature	Family	Definition	Window	Upstream fields	Write time	Fairness note
<code>recency_days</code>	RFM	Days since customer’s most recent prior transaction.	All prior transactions	<code>customer_id</code> , <code>ts</code>	<code>ts</code> of each transaction	May reflect differences in access to or use of financial services.
<code>frequency</code>	RFM	Number of prior transactions for the customer.	All prior transactions	<code>customer_id</code> , <code>txn_id</code> , <code>ts</code>	Transaction creation time	Higher activity may reflect greater service access.
<code>monetary_in</code>	RFM	Total incoming transaction value for the customer.	All prior transactions	<code>customer_id</code> , <code>amount_abs</code> , <code>amount_signed</code> , <code>txn_type</code> , <code>ts</code>	Transaction posting time	May reflect economic capacity or income patterns.
<code>monetary_out</code>	RFM	Total outgoing transaction value for the customer.	All prior transactions	<code>customer_id</code> , <code>amount_abs</code> , <code>amount_signed</code> , <code>txn_type</code> , <code>ts</code>	Transaction posting time	May proxy socioeconomic circumstances or expenditure capacity.
<code>cashout_ratio</code>	Ratios	Proportion of observed transactions classified as cash-out activity.	All prior transactions	<code>customer_id</code> , <code>txn_type</code> , <code>ts</code>	Transaction event time	May differ across underserved segments.

Feature	Family	Definition	Window	Upstream fields	Write time	Fairness note
out_in_ratio	Ratios	Outgoing transaction value relative to incoming transaction value.	All prior transactions	monetary_out, monetary_in	Derived after source aggregates are available	May encode differences in economic circumstances; segment-level calibration should be monitored.
n_counterparty	Stability	Number of distinct counterparties observed for the customer.	All prior transactions	customer_id, counterparty, ts	Transaction event time	May reflect business type or market access.
gps_missing	Behavioural	Indicator that GPS latitude is unavailable.	Event time	gps_lat	GPS availability at event time	May proxy geographic or network coverage disparities
payday_dist	Behavioural	Distance from nearest 1st/28th payday anchor.	Current transaction	ts	Transaction event time	Calendar behaviour may vary by livelihood.
hr_sin	Behavioural	Sine encoding of transaction hour.	Current transaction	ts	Transaction event time	Transaction timing may correlate with occupational or socioeconomic patterns.
hr_cos	Behavioural	Cosine transformation of the transaction hour to represent cyclical time-of-day behaviour.	Current transaction	ts	Transaction event time	Same fairness consideration as the corresponding time-of-day signal.
dow_sin	Behavioural	Sine transformation of day-of-week to represent weekly transaction cycles.	Current transaction	ts	Transaction event time	Weekly transaction patterns may differ across employment and livelihood groups.
dow_cos	Behavioural	Cosine transformation of day-of-week to represent weekly transaction cycles.	Current transaction	ts	Transaction event time	Weekly transaction patterns may act as indirect socioeconomic signals.
amount_abs	RFM	Absolute transaction value after standardizing the signed amount representation.	Current transaction	amount, amount_signed	Transaction posting time	It is an economic signal and may reflect differences in product use or economic circumstances.
log_amt	RFM	Log-transformed absolute transaction amount used to reduce skew.	Current transaction	amount_abs	Available immediately after transaction amount is recorded	Retains the socioeconomic information present in transaction value.

3 Leakage Assessment Memo

TO: Head of Credit Risk

FROM: Hilda Ndambuki, Data Science / Credit Risk Analytics

DATE: 15/08/2026

SUBJECT: Data Leakage Assessment – Mobile-Money Credit Risk Model

Purpose. This memo assesses the two intentionally planted sources of data leakage in the mobile-money fraud-risk modeling pipeline. Leakage was evaluated by examining the affected variables, their relationship with the fraud outcome and the change in ROC-AUC after the leakage sources were removed. Both variables were excluded from the final predictive feature set because they contain information generated after outcome resolution process.

Leakage Source 1: `manual_review_score`

Location: `manual_review_score`.

The first leakage source is `manual_review_score`. The correlation analysis between this variable and the fraud label shows an approximately 0.98 correlation. The variable represents information generated through the manual review process and therefore would not be legitimately available as a predictor at the intended scoring point. This corresponds to the **target leakage** mechanism. The leakage occurs because information associated with the business outcome is included in the same feature table used for learning. The model can therefore use information generated after the event being predicted.

The impact was substantial. The leakage-free model achieved a ROC-AUC of approximately 0.625, whereas inclusion of `manual_review_score` increased it to approximately 1.000. This represents an absolute increase of approximately 0.375 ROC-AUC.

Governance recommendation: Post-outcome review variables should be excluded from candidate training datasets. A mandatory feature sign-off process should document the source system, write time, data owner and scoring availability of every candidate feature. Features that fail the as-of availability check should be rejected before model training.

Leakage Source 2: `settlement_status`

Location: `settlement_status`.

The second source of leakage is `settlement_status`, which records the resolution state of a transaction. The analysis showed strong target like separation across its categories: transactions with status **reversed** are 100% fraud, transactions with status **held** are approximately 65% fraud, while the remaining statuses have fraud rates close to zero. This pattern indicates that the variable contains information closely related to the eventual outcome. This is also a **target leakage** mechanism. The field becomes informative through the downstream resolution process and therefore represents future information relative to the original scoring decision. Including it allows the model to use transaction outcome rather than solely the information available when the credit-risk decision is made.

The impact was again substantial. The leakage free model achieved a cross-validated ROC-AUC of approximately 0.625, compared with essentially 1.000 when `settlement_status` was included. This corresponds to an absolute increase of approximately 0.375 ROC-AUC. The result confirms that the apparent predictive contribution of the variable is dominated by information that would not be available at the intended scoring point.

Governance recommendation: Settlement and case management outputs should be maintained in a post decision data layer and prevented from being automatically merged into modeling datasets. Candidate feature lists should also be validated against an approved upstream data lineage register before model training.

4 Reproducible Machine Learning Pipeline

The final predictive model was implemented as a single end-to-end Scikit-learn **Pipeline** beginning from the validated engineered features. This was completed upstream using the as-of rule, with all customer level aggregates constructed only from transactions occurring before the scoring timestamp. The resulting modeling table contains 15 numerical features and three categorical features. The numerical preprocessor consists of median imputation with missingness indicators, Yeo–Johnson transformation and robust scaling. Categorical variables are imputed using an “UNK” category and one-hot encoded, with infrequent categories handled by the encoder. Logistic Regression is then fitted as the final estimator. Both identified post-outcome leakage variables, `manual_review_score` and `settlement_status`, are excluded from the final feature set.

4.1 Group-Aware Evaluation

The model was evaluated using five-fold **GroupKFold** cross-validation, with the resolved `customer_id` used as the grouping variable. This prevents transactions belonging to the same customer from being distributed between the training and validation folds. The final model achieved the following mean group-aware cross-validated performance:

Table 3: Group-aware cross-validated performance

Metric	Cross-validated result
ROC-AUC	0.6250
PR-AUC	0.1349

The ROC-AUC value indicates modest discrimination between fraudulent and non-fraudulent transactions. The corresponding PR-AUC is relevant for this application because fraud is the minority outcome.

4.2 Pipeline Persistence and Reload Verification

After evaluation, the final pipeline was fitted on the complete feature engineered modeling dataset and saved using `joblib` to `credit_risk_pipeline.joblib`. The saved object was then reloaded and executed successfully, confirming that it contains the required components for data transformations and model components. The pipeline expects a feature engineered input, while all standardization and algorithmic encoding required by the predictive model is performed internally.

Conclusion

The data quality audit provides an evidence base for remediation, while the engineered features incorporate domain knowledge from the five feature families. The leakage assessment demonstrates the importance of enforcing the as-of principle and controlling information flow during model development. Finally, the use of a single Scikit-learn pipeline and group aware cross-validation provides a reproducible framework in which preprocessing and model training can be independently repeated.